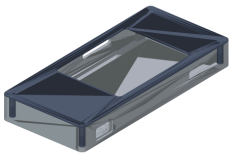
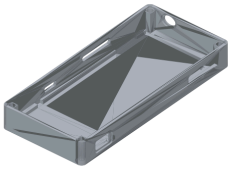
Pod case, Rev B

Two printed parts, **screwed together from underneath**: a lower **tray** that carries the perfboard and a flat **bezel** that frames the glass. Four M2 screws go up through the floor into pegs on the bezel; the heads are countersunk and sit under the rubber feet, so nothing shows on the outside. Outer size **57.2 × 125.0 × 21.8 mm** (Rev A was 57.2 × 114.2 × 22.7).

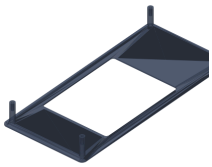
Assembled



Lower tray, from above (Pico end bottom-right)



Upper bezel, turned over (inside)



What changed from Rev A, and why

- **Screws instead of snaps.** The perfboard fills the whole cavity, so there is nowhere beside it for screw posts. The case is therefore 5.4 mm longer at each end: that strip holds a screw block in each corner. The bezel's pegs drop into those blocks and the screws come in from the floor. Truly internal screws in a two-piece sandwich are unreachable once it is closed; from-below under the feet is the hidden version that can actually be assembled.
- **Slimmer.** Floor 1.6, walls 1.8, 3.0 mm wire clearance, bezel plate 1.5: 21.8 mm total. The 8.5 mm screen sockets are what sets the height; everything else is at its minimum.
- **Round jack hole**, Ø8.2, fully inside the tray wall, with an inside counterbore so the jack barrel can enter the wall.
- **Cavities sized from the ordered parts** (table below). The TP4056 moved to the right side under the screen, because the top end is full (DAC, SD socket) and a USB-C socket must sit within ~1 mm of the outer surface to take a plug.
- **Video re-analysis:** the screen module ends ~8 mm before the top edge of the perfboard (jack and SD socket are beyond it), so the window moved down to centre y 60 (was 66.5); the Pico sits flat on the board, so its micro-USB opening dropped to z 4.5–12.5; the DAC sits on a ~4 mm standoff, jack centre 8 mm above the board.

Parts and the spaces made for them

Part (ordered)	Size used	Where it goes (board coords)	Opening / clearance
TP4056 Type-C charger with protection	29 × 17.3 × 4.3	on the board under the screen, x 34.7–52, y 74–103, socket facing right, PCB overhanging the board edge by 0.8	right wall slot 11 × 7, z 9.8 centre, 1 mm outside recess (14 × 10)
Right-angle mini slide switch	body 13 × 6.5 × 5.5, 2.54 pitch	board's left edge, x 0–6.5, y 24.5–37.5, lever out the left wall	left wall slot 10 × 4.5, z 7.0–11.5
LiPo 403048 (or KP-403450)	30 × 48 × 4 (34 × 50 × 4)	on the board under the screen, x 19–49, y 25–73, foam tape	8.5 mm headroom under the screen
GY-PCM5102 DAC	≈31 × 17.5 (jack end)	top-left, x 3.8–21.3, jack end 3.5 mm past the board edge, PCB ≈4 mm above the board	round hole Ø8.2 in the top wall, centre x 12.5, 8 mm above the board
Pico 2 W (fitted)	51 × 21, flat on the board	y −0.3–20.7, USB on the right	right wall slot 12 × 8, z 4.5–12.5, 1 mm recess
Screen 2.8" (fitted)	50 × 86, glass top 14.1 above the board	x 1–51, y 15–101; SD holder underneath, left	window 46 × 60 at (26, 60); left wall slot 24 × 4.5 for the card
1N5819, 100 nF	—	on the board as in the build guide	—

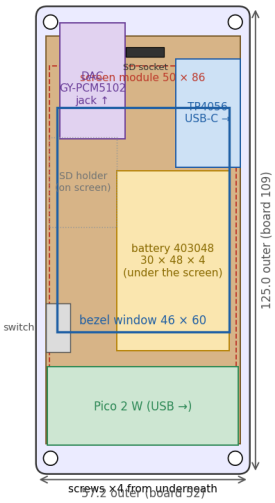
Sources: TP4056 Type-C 29 × 17.3 × 4.3 (AZ-Delivery datasheet); switch 13 × 6.5 × 5.5 (makerbazar listing); 403048 = 4.0 × 30 × 48 by model number; GY-PCM5102 size is from typical listings (not on the makerbazar/robu pages, which give no dimensions), hence the Ø8.2 hole is generous and the hole position is a parameter (*JACK_X, JACK_H*).

Screws and feet

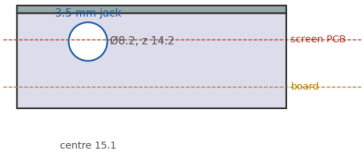
- 4 × **M2 × 8 self-tapping** (or M2 × 8 machine screws run into the 1.6 mm pilot: PLA takes M2 directly), countersunk heads.
- 4 × **Ø8 stick-on rubber feet**, 1–2 mm thick, pressed into the Ø7 × 0.5 recess over each screw head.
- Pegs on the bezel: Ø3.4, 18.2 long, in Ø3.8 bores (0.2 mm per side). Print the bezel face-down; the pegs grow straight up, no supports.

Dimensions (mm)

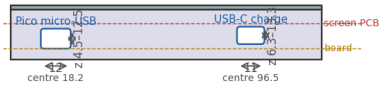
Plan, looking into the tray (board coordinates)



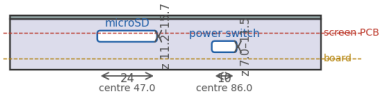
Top end wall (jack), from outside



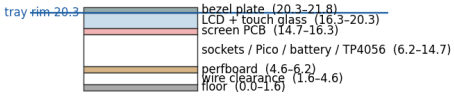
Right wall (Pico USB, USB-C charge), from outside



Left wall (switch, microSD), from outside (mirror of plan)



Stack (section), mm from the outside of the floor



Key numbers

```
Outer: 57.2 × 125.0 × 21.8 mm, corner radius 3
Walls 1.8, floor 1.6, bezel plate 1.5
Board clearance 0.8 per side, 5.4 at each end
Wire clearance under the board 3
Tray rim at the glass top (20.3); bezel lip 1.0 × 1.2
Four screw blocks in the corners, bore Ø3.8;
bezel pegs Ø3.4, pilot Ø1.6 × 7 for M2 × 8
self-tapping screws from below, countersunk,
covered by Ø8 rubber feet (recess Ø7 × 0.5)
Jack hole Ø8.2, centre 8 above the perfboard
Window 46 × 60, centre (26, 60) board coords
USB ports: 1.0 mm outside recess so plugs seat fully
```

Before printing: three checks

1. **Window:** print the 1:1 template (last page, 100 % scale) and lay it on the device with the dashed board outline on the perfboard. The blue window must frame the touch panel with 1–2 mm of glass hidden on every side. The screen's position was read from the video, so this is the number most likely to need a nudge: tell me the shift and I move *WIN_CY*.
2. **Jack height:** when the new DAC is in, measure perfboard top → jack centre. The model assumes 8 mm; the Ø8.2 hole tolerates ±1.5 mm. If you mount it higher, say so.
3. **Under-board wires:** there is 3.0 mm under the board. Press the yellow loops flat and trim the long leads, or raise *UNDER* to 4.

Fitting the new parts

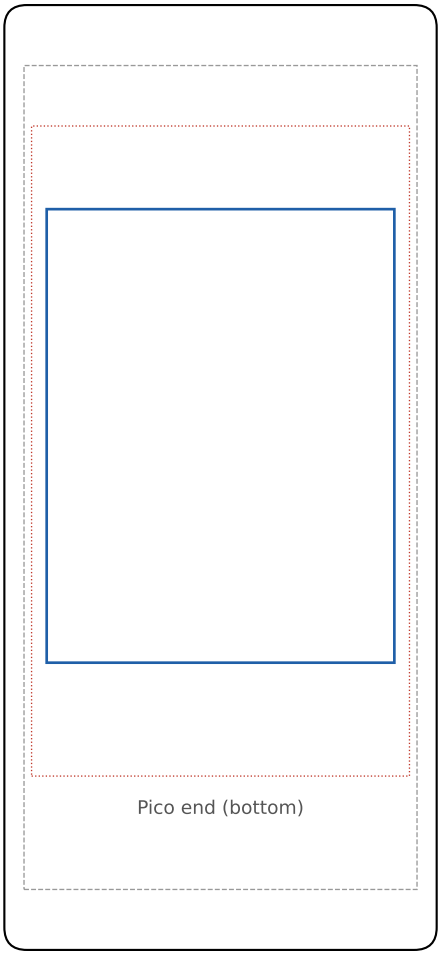
- **DAC:** on short pins (≈4 mm standoff) at x 3.8–21.3 with the jack end 3.5 mm past the board's top edge; the corner screw block is to its left, the SD socket to its right. SCK pad to GND on the generic board.
- **TP4056:** foam tape on the board at x 34.7–52, y 74–103, USB-C socket facing the right wall, hanging 0.8 mm over the board edge (it will touch the wall's inner face). OUT+ → switch → 1N5819 → VSYS, OUT− → GND, B+/B− to the battery.
- **Switch:** body against the board's left edge at y 24.5–37.5, lever through the slot; two wires to the power path.
- **Battery:** foam tape, x 19–49, y 25–73, wires toward the TP4056.

Print settings and assembly

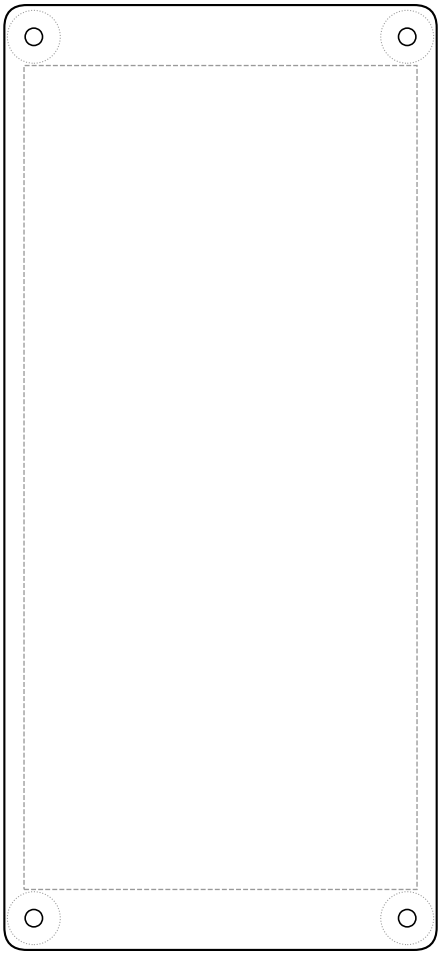
- PLA or PETG, 0.2 mm layers, 3 walls, 25 % infill, 100 % infill for the bezel (thin plate + pegs). Tray prints open side up, bezel face down. No supports.
- Drop the board in, jack end first (the jack goes into its counterbore), then the Pico USB into its slot. Press the bezel on: the lip locates it, the pegs enter the corner blocks. Turn over, four screws, four feet.
- Tolerances built in: 0.8 mm around the board, 5.4 mm at the ends, 0.2 mm on the lip and pegs. If the pegs are tight, run a 3.8 mm drill through the bores.

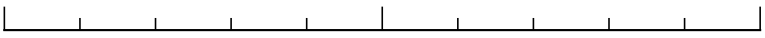
Files: *pod_tray.stl/.step*, *pod_bezel.stl/.step*, *pod_case_assembled.step*, source *pod_case.py* (every number above is a named parameter at the top). STEP opens directly in Fusion 360.

BEZEL (top)



FLOOR (underneath, mirrored) with screw holes





100 mm check line

Pod case Rev B, 1:1. Print at 100 %. Lay the bezel outline on the device with the dashed board outline on the perfboard: the blue window must frame the touch panel with 1–2 mm of glass hidden under the bezel on every side.